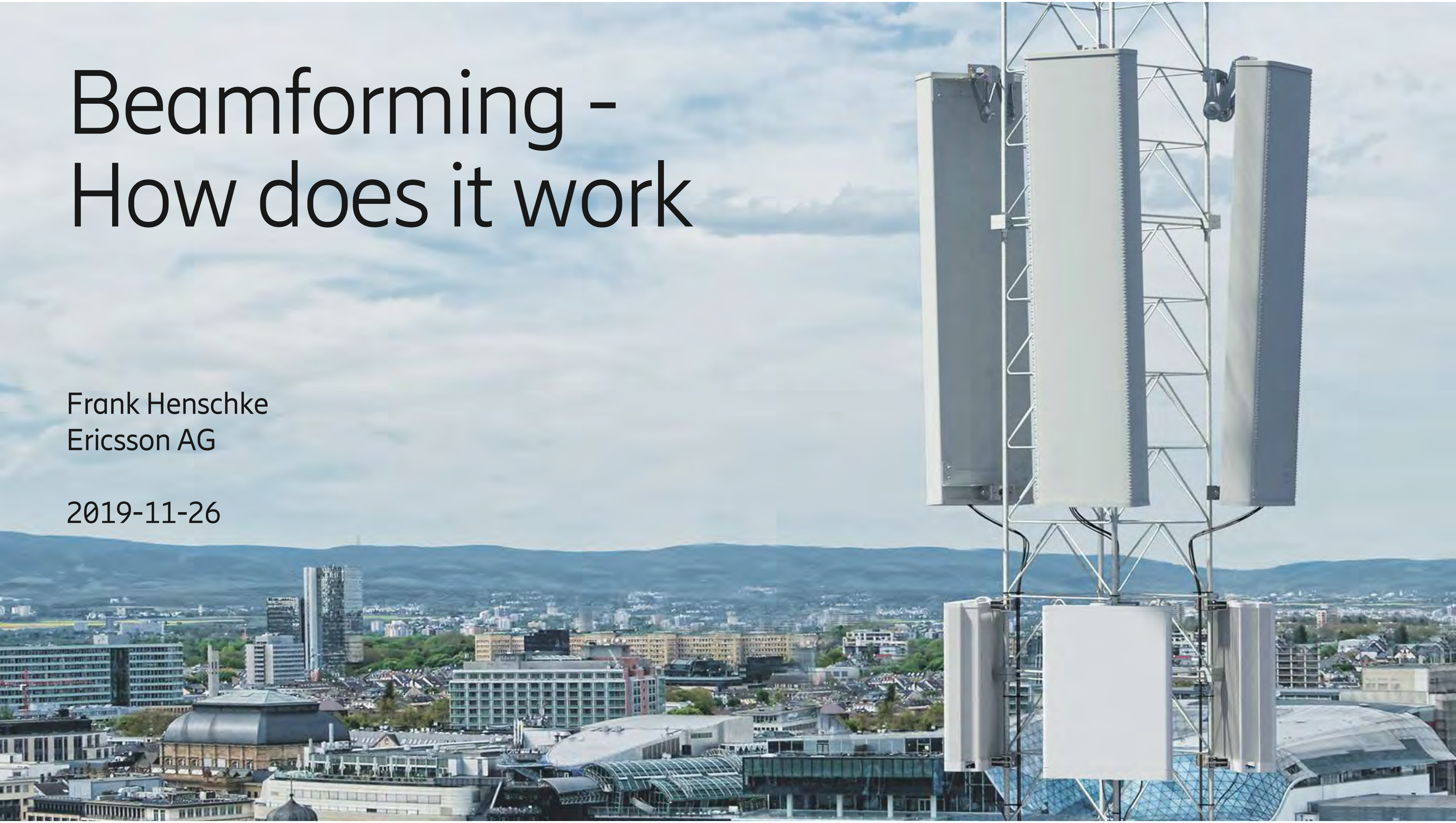
Beamforming - How does it work

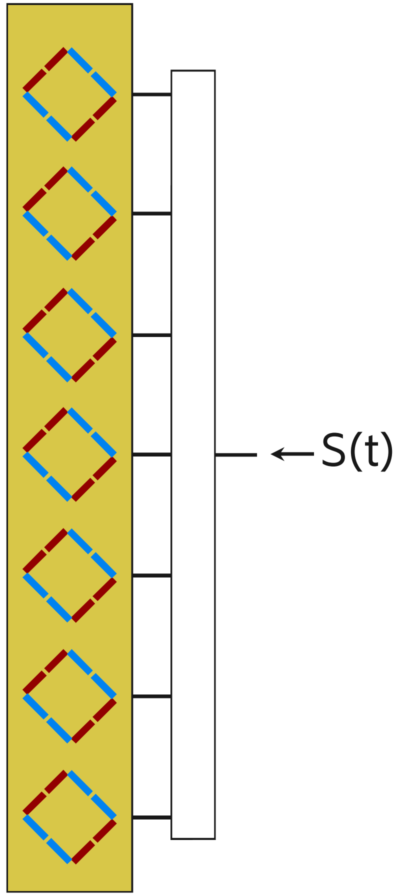
Frank Henschke
Ericsson AG

2019-11-26

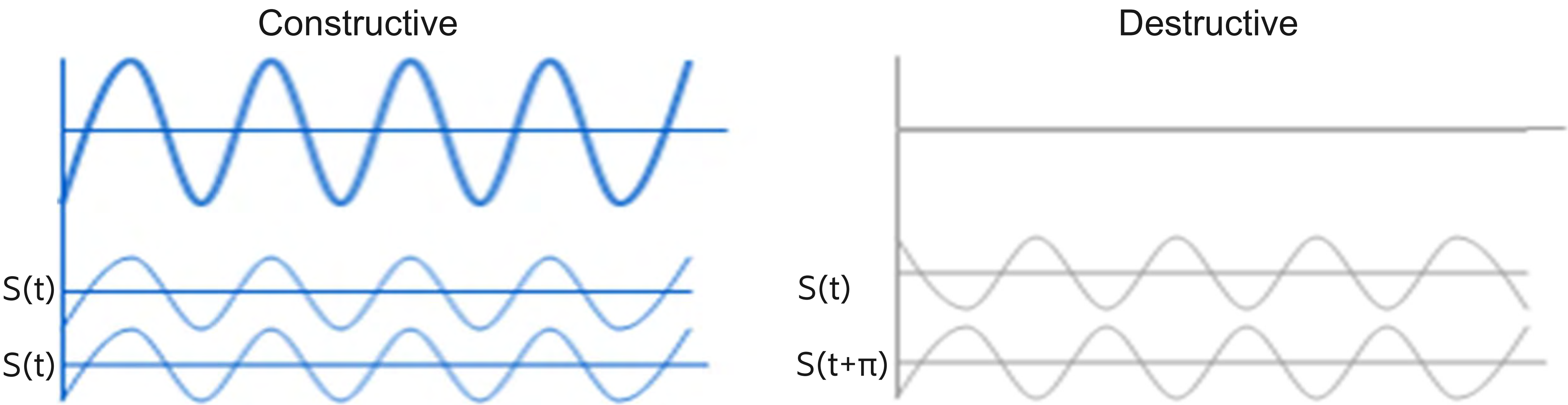


Antenna, so far…

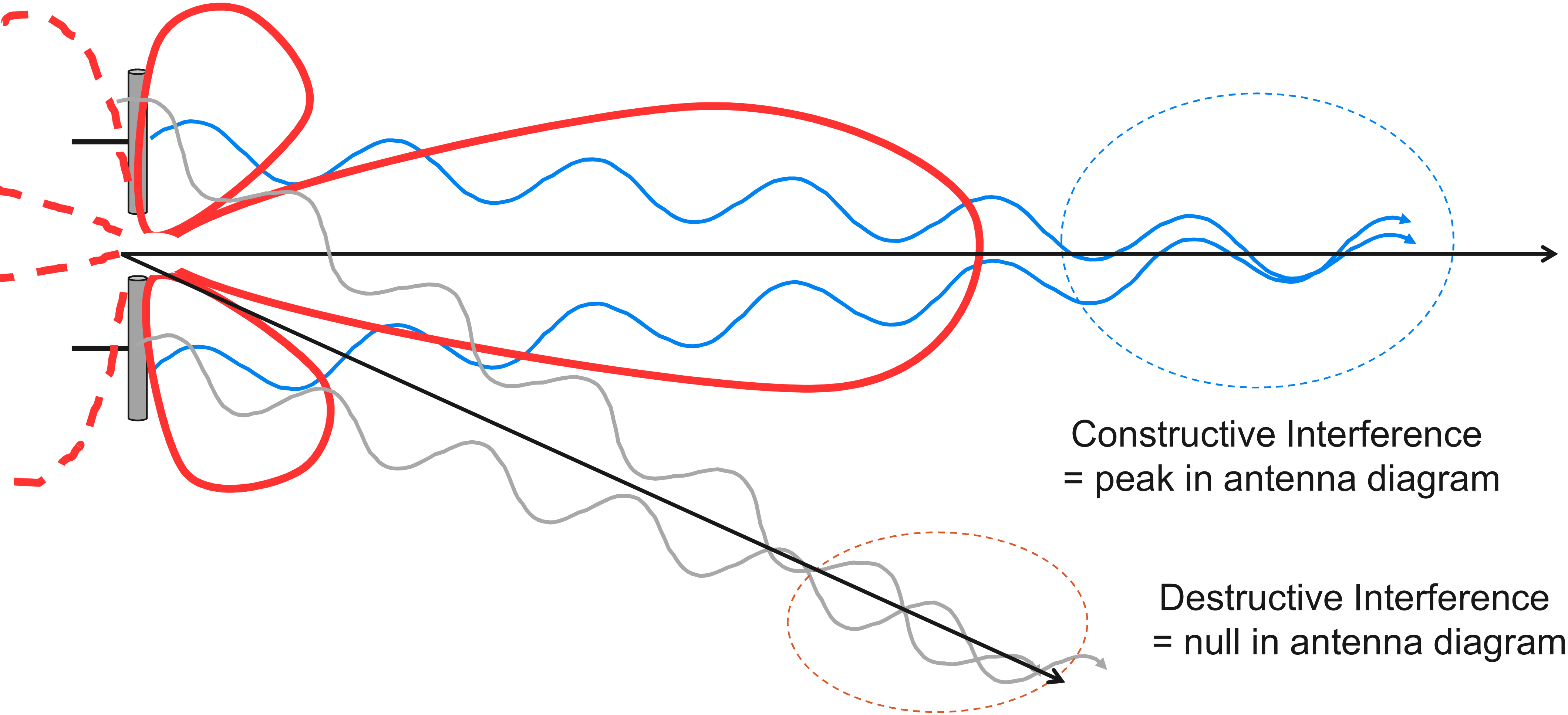
- Antenna are provided with antenna diagrams (“beams”)
- Each antenna has its own beam diagram
- Antenna with Electrical tilt are provided with multiple beam diagrams, one for each tilt
- But what is behind beam shape and beam direction?



Interaction between electromagnetic waves

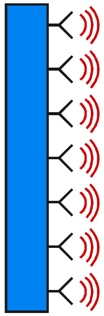


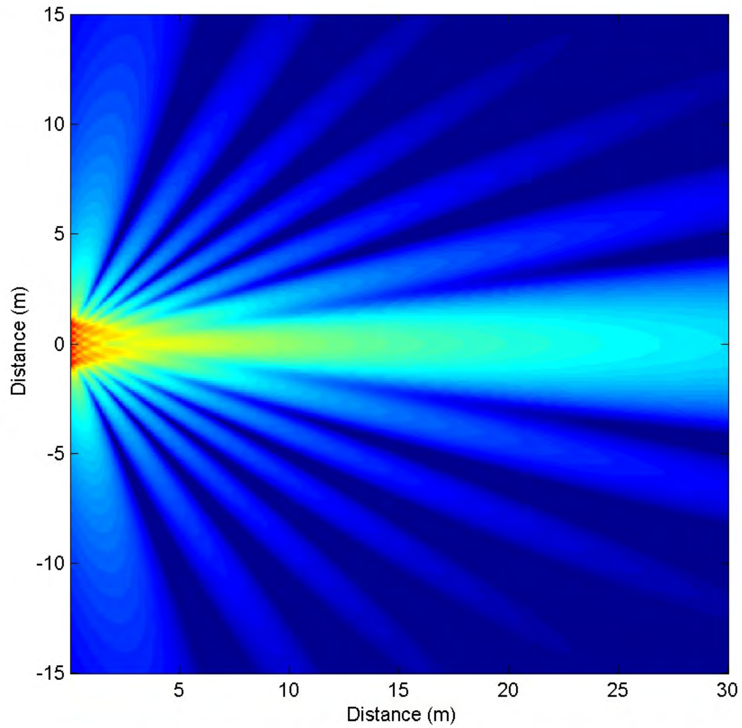
Directional Properties

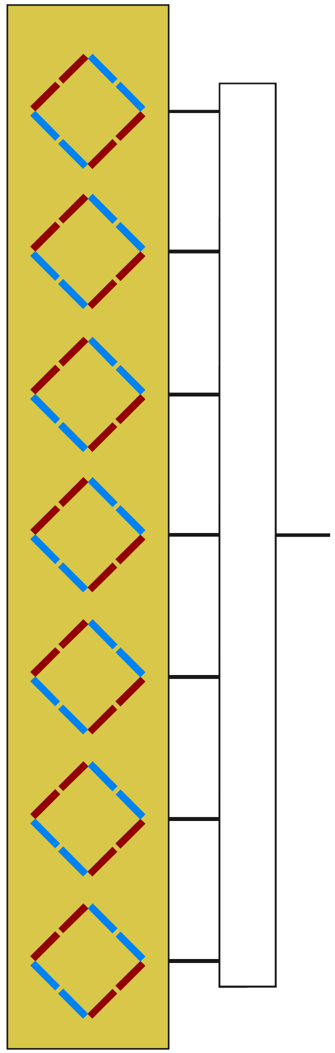


Behind the beam shape

- The higher the number of elements:
  - The narrower the beam
  - The higher the antenna gain

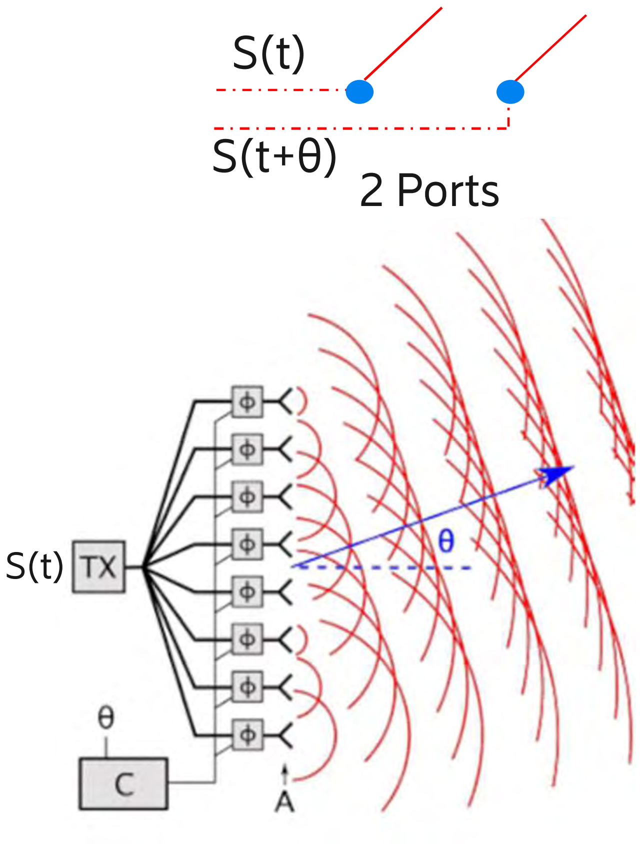


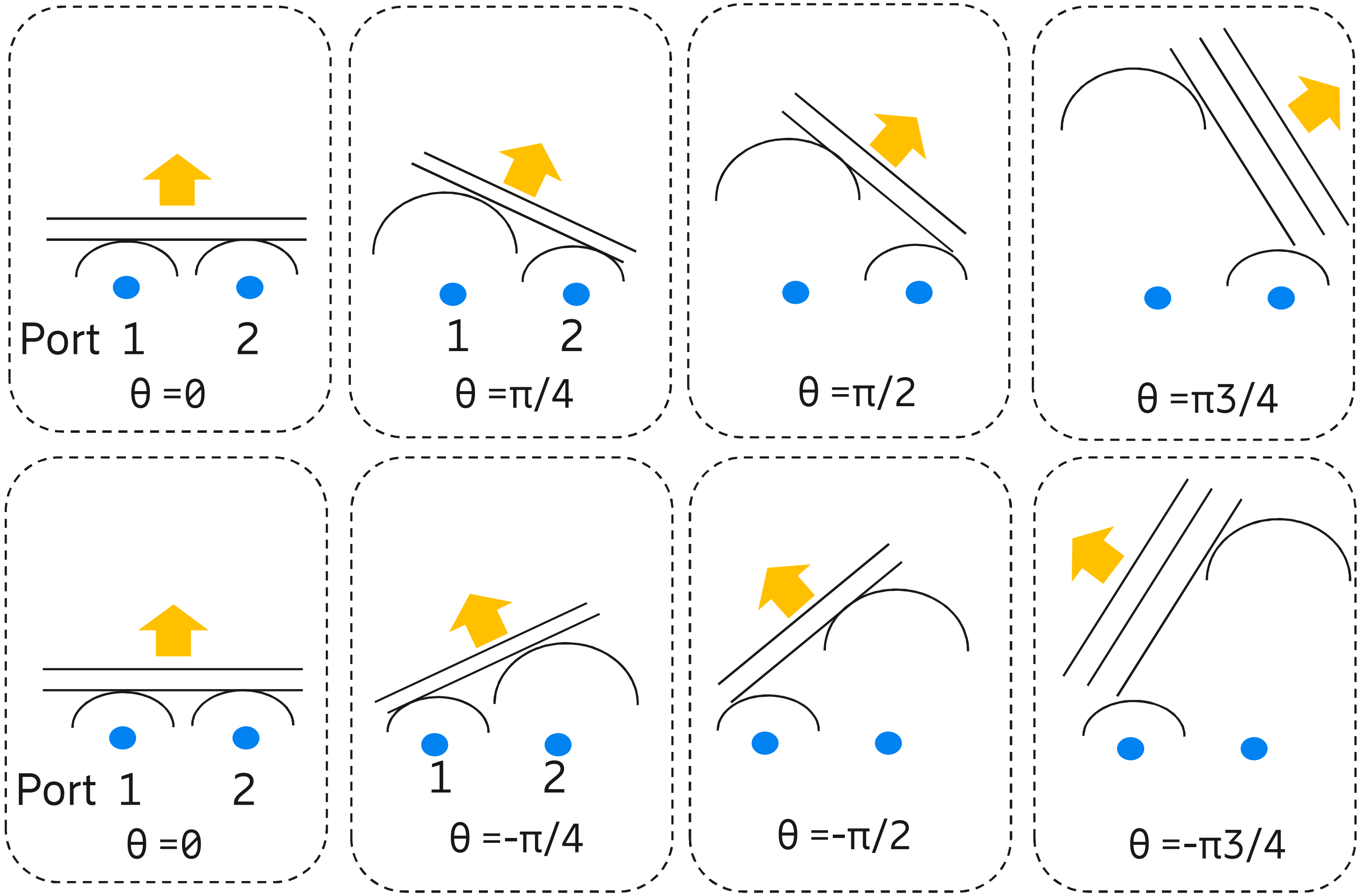




Behind the beam direction

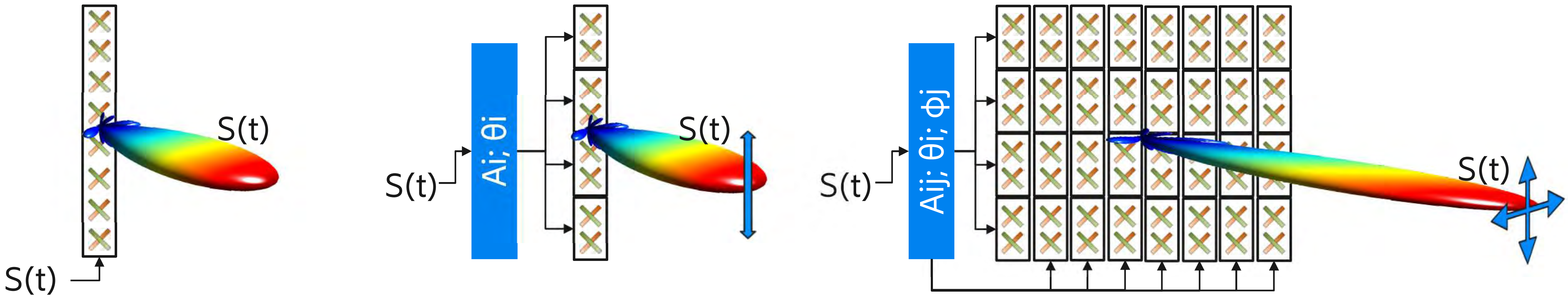
Phase shift between antenna dipoles defines the direction of the beam





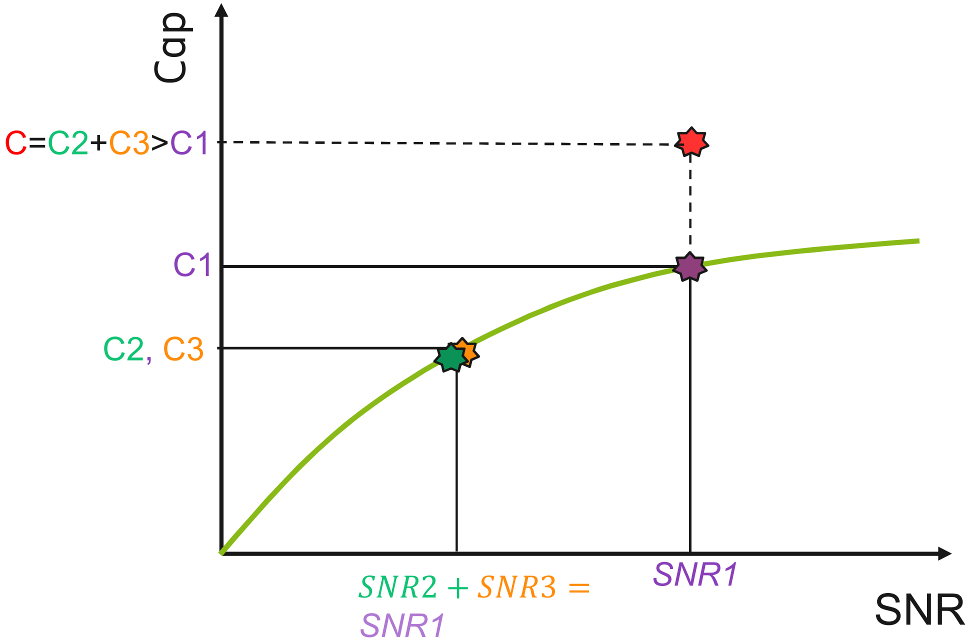
Beamforming

- By splitting the antenna in subarrays and sending the same signal with proper amplitudes and phases to the different subarrays it is possible to dynamically change both beam shape and beam direction
- With beamforming both amplitude and phase are part of the baseband signal processing in the radio node => the radio node decides both beam shape and beam direction
- By placing multiple arrays of subarrays next to each other the radio node can steer both beam shape and beam direction in both vertical and horizontal domains



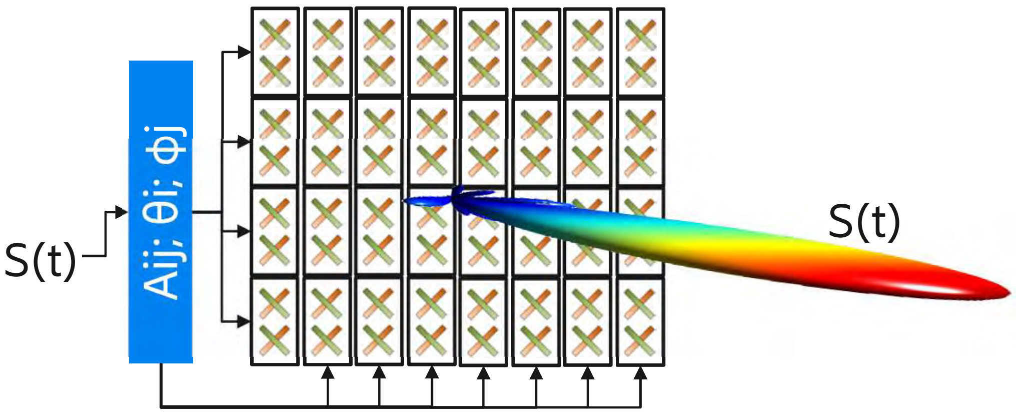
MIMO

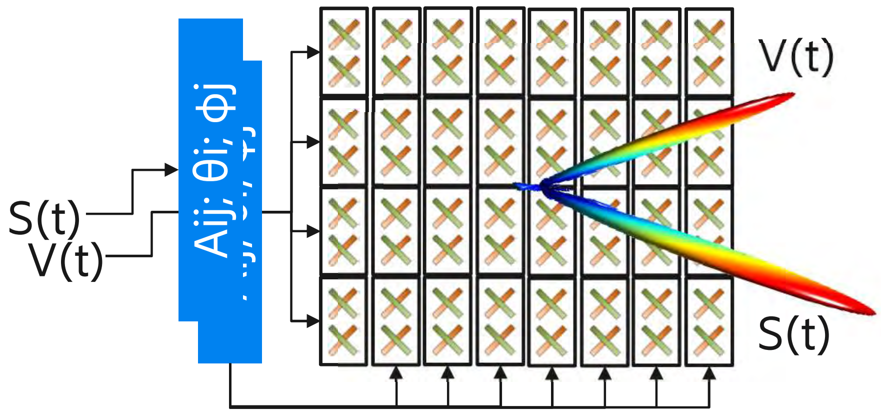
- According to Shannon formula there is limited gain in channel capacity when SNR is high
- At high SNR it is more efficient to send 2 data streams over channels with lower SNR than sending a single data stream over a channel having very good SNR



Beamforming & MIMO

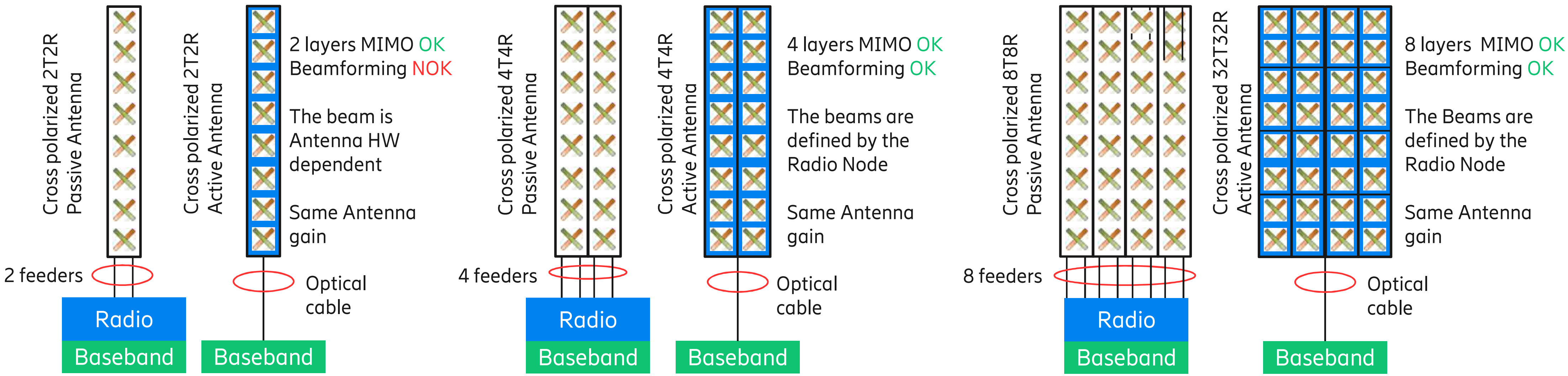
- At high SNR the radio node would split power among different data streams in order to exploit MIMO benefits
- multiple data streams are sent to the antenna, each data stream uses its own channel and a dedicated set of amplitudes and phases (=beam)
- each beam is emitted with lower power
- SU-MIMO: multiple data stream are sent to a single user for increased user bitrate
- MU-MIMO: multiple data streams are sent towards multiple users for increased system capacity





Definitions and considerations

- Active antenna = passive antenna with integrated electronics such as signal amplifiers, analog to digital converters and other signal processing units
- Beamforming is achieved when the same signal is sent to different (correlated) elements of the antenna with well defined phases and amplitudes
  - Beamforming can be achieved with both passive and active antenna, passive antenna with multiple ports are required to perform beamforming



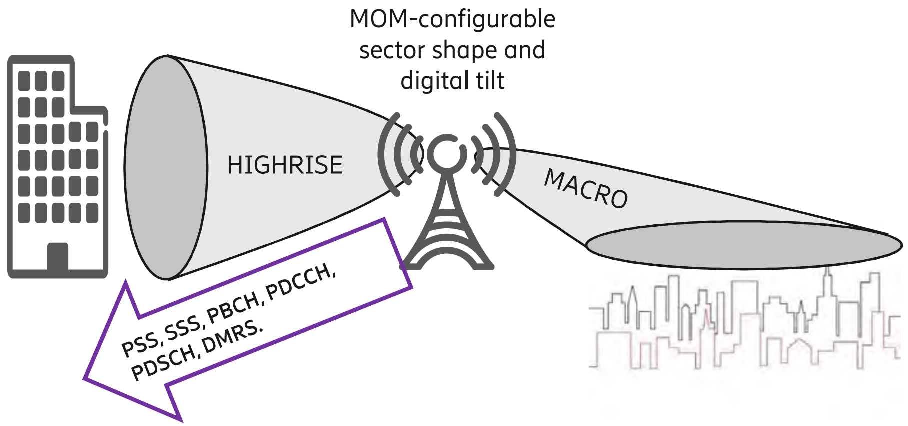
Broadcast Beams and Traffic Beams

Beamforming capable antenna

- There are 2 types of beams:
  - Broadcast beam for common channels, always on air, beam is defined by the radio node
  - Traffic beams for user data, only on air when there is data to be sent, this is a high gain narrow beam

- Common channels can be sent via multiple narrow beams or a single wide beam, Ericsson implementation uses a single wide broadcast beam: beamforming is in this case used to achieve the wanted cell coverage, tilt may apply.
- Broadcast beam has lower ERP than traffic beams

  - Four options available for broadcast beams:
    - Macro (providing standard Macro coverage with 60° Horizontal, 10° Vertical)
    - Hotspot (providing coverage to a close area: 60° Horizontal, 30° Vertical)
    - Highrise (providing coverage to a close high rise building with 20°Horizontal and 30° vertical)
    - Customized

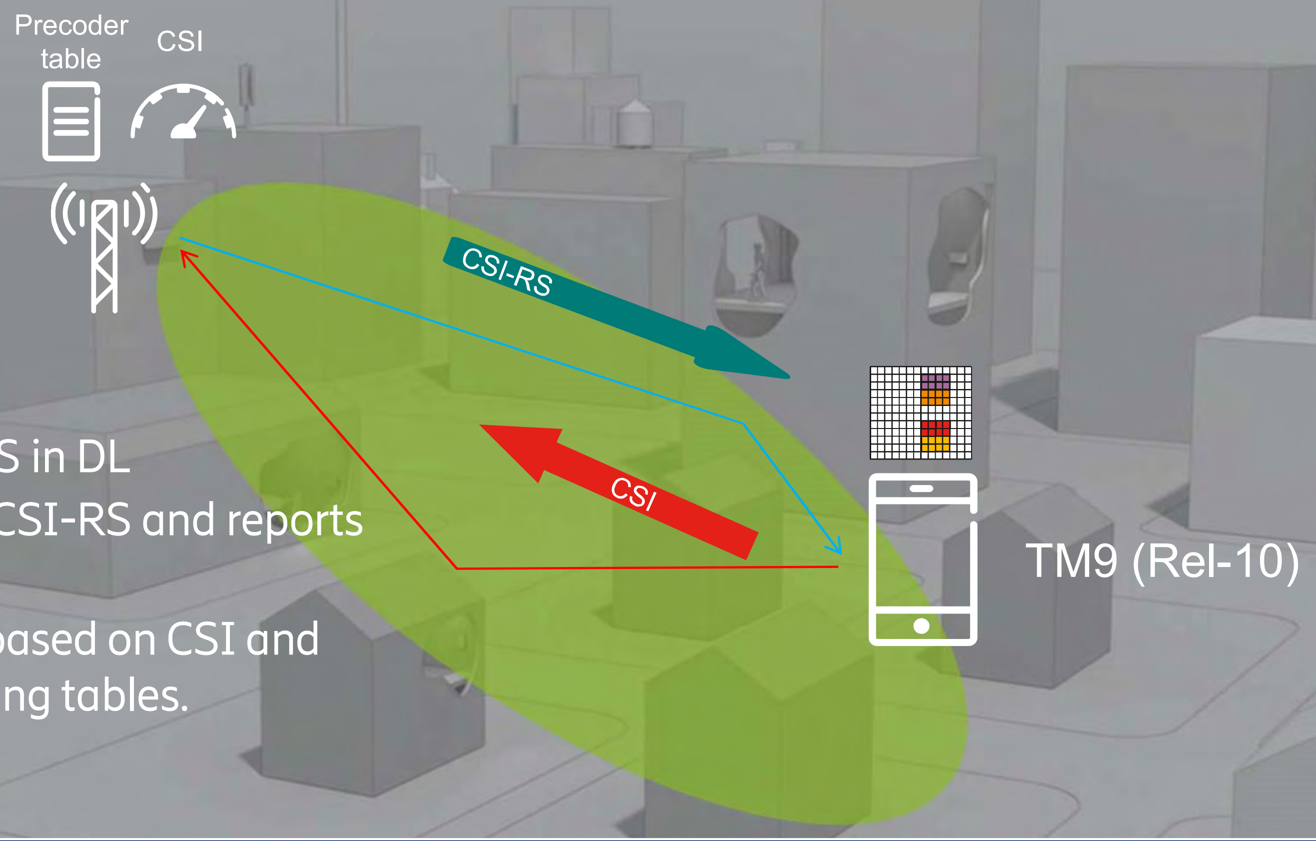


Traffic beams: who decides what beam is on air?

- For Traffic beams there are 2 main types of beamforming:
  - Reciprocity based beamforming
    - The User Equipment (UE) transmits reference signals and the radio node performs channel estimations in order to define what direction, how many beams to use and which shape
    - For SU-MIMO the radio can use up to as many channels/beams as many transmitters are available on the User equipment
    - For MU-MIMO The radio can theoretically use up to as many channels/beams as many transmitters are available on the Radio
  - Codebook based beamforming
    - The radio node transmits reference signals and the User Equipment (UE) performs channel estimations in order to requests specific beam directions to the radio node, the UE can select among a list of 3GPP predefined beam directions (Precoding codebook)
    - The list of predefined beam directions is configuration dependent, it depends on how many antenna ports/CSI-RS the radio node is configured to transmit
    - Any tilt configured on broadcast beams would also affect the traffic beams
  - The radio node could forbid utilization of beams in specific directions (codebook restriction for codebook basedbeamforming) or create nulls in specific directions (reciprocity based beamforming)

Closed-loop CSI feedback

UE specific beamforming, LTE example

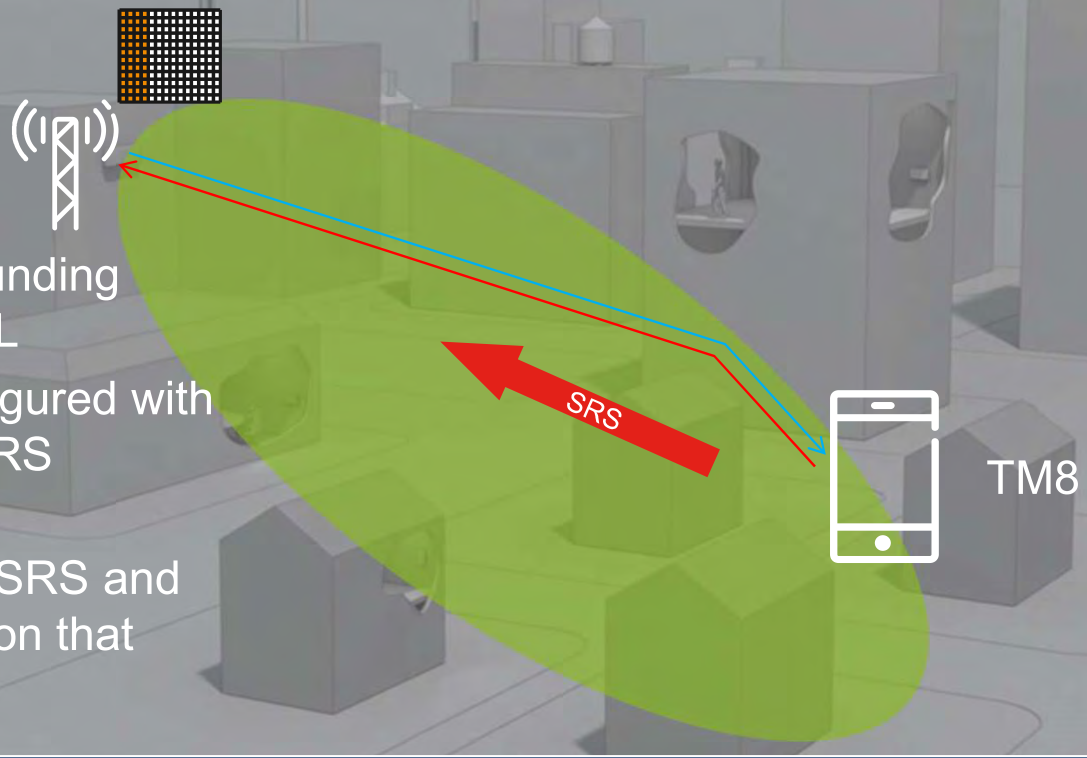


- eNB transmits CSI-RS in DL
- UE measures on the CSI-RS and reports back CSI
- DL beam forming is based on CSI and standardized precoding tables.

Works for both FDD and TDD

Reciprocity based CSI
UE specific beamforming, LTE example

- UE transmits SRS, sounding reference signals, in UL
- Different UEs are configured with different UE specific SRS configuration
- eNB measures on the SRS and base DL transmission on that

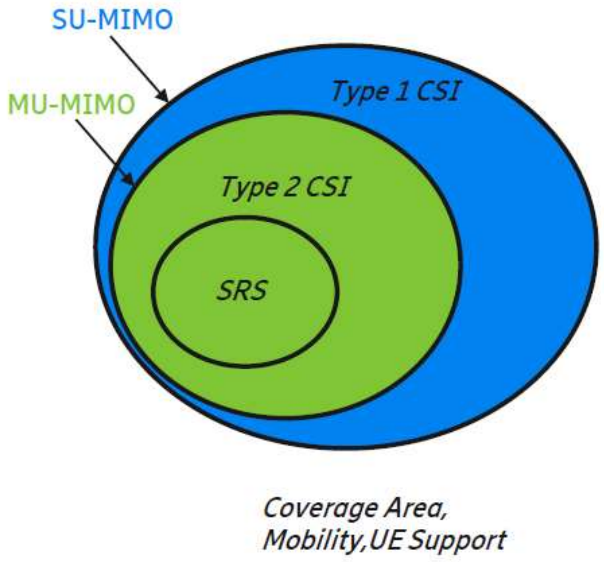


Works for TDD but not so good for FDD

Traffic beams: who decides what beam is on air?

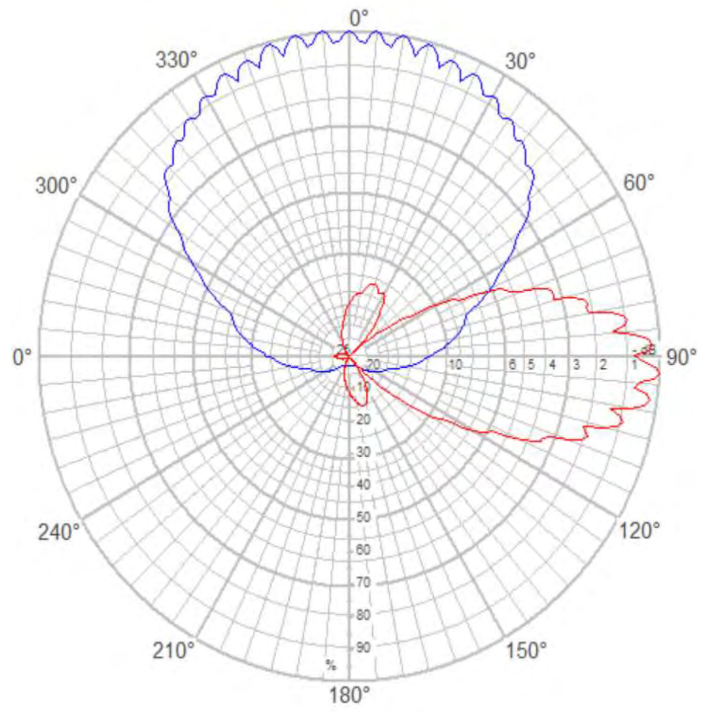
- Reciprocity based beamforming via Uplink channel sounding
  - UE transmits *Sounding Reference Signals (SRS)*
  - The radio node measures and defines what and how many beams to send

- Codebook based beamforming via downlink reference signals, so called "Channel State Information Reference Signals" (CSI-RS)
  1. Type 1 CSI codebook:
     - *Radio node sends CSI-RS signals according to 3GPP, the UE measures and asks for specific beams out of 3GPP defined codebooks*
       - *CSI-RS can be pre-coded, non pre-coded*
       - *Suitable for SU-MIMO*
  2. Type 2 CSI codebook
     - *High resolution CSI feedback targeting MU-MIMO*



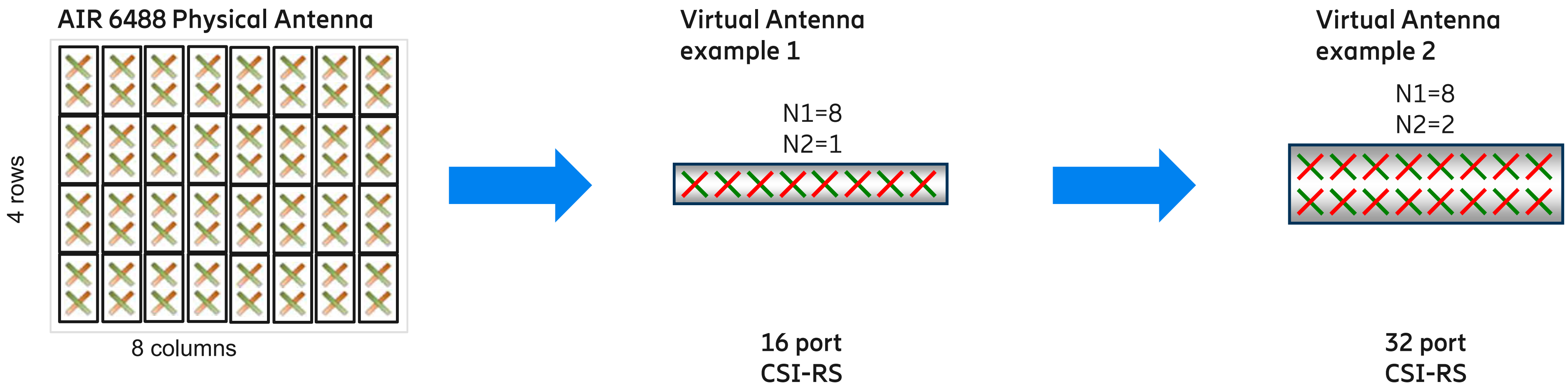
Antenna patterns for reciprocity based beamforming
64T64R

- With reciprocity based beamforming the radio node has access to all TX ports
- As a consequence there is higher flexibility to shape beams and point the traffic beam in a wider range of directions
- The picture shows envelope of traffic beams
- Any tilt on the broadcast beam would not have any effect on the envelope of traffic beams



Antenna patterns for codebook based beamforming

64T64R



- In codebook based beamforming the UE sees as many antenna ports as configured by the radio node through CSI-RS
- each configuration (N1, N2) corresponds to a 3GPP defined amount of possible beams

CSI-RS configuration and number of available beams

- Based on 38.214 v15.3-Table 5.2.2.2.1-2: Supported configurations of (N1,N2) and (O1,O2).

Number of beams in the horizontal domain

Number of beams in the vertical domain

Number of CSI-RS antenna ports	(N1,N2)	2*N1*N2 = # of CSI ports	(O1,O2)	N1*O1	N2*O2	N1*O1*N2*O2=Ac (Bitmap in SinglePanel n1-n2)
4	(2,1)	2*2*1 = 4	(4,1)	2*4=8	1*1=1	2*4*1*1 = 8
8	(2,2)	2*2*2 = 8	(4,4)	2*4=8	2*4=8	2*4*2*4 = 64
	(4,1)	2*4*1 = 8	(4,1)	4*4=16	1*1=1	4*4*1*1 = 16
12	(3,2)	2*3*2 = 12	(4,4)	3*4=12	2*4=8	3*4*2*4 = 96
	(6,1)	2*6*1 = 12	(4,1)	6*4=24	1*1=1	6*4*1*1 = 24
16	(4,2)	2*4*2 = 16	(4,4)	4*4=16	2*4=8	4*4*2*4 = 128
	(8,1)	2*8*1 = 16	(4,1)	8*4=32	1*1=1	8*4*1*1 = 32
24	(4,3)	2*4*3 = 24	(4,4)	4*4=16	3*4=12	4*4*3*4 = 192
	(6,2)	2*6*2 = 24	(4,4)	6*4=24	2*4=8	6*4*2*4 = 192
	(12,1)	2*12*1 = 24	(4,1)	12*4= 48	1*1=1	12*4*1*1 = 48
32	(4,4)	2*4*4 = 32	(4,4)	4*4=16	4*4=16	4*4*4*4 = 256
	(8,2)	2*8*2 = 32	(4,4)	8*4=32	2*4=8	8*4*2*4 = 256
	(16,1)	2*16*1 = 32	(4,1)	16*4= 64	1*1=1	16*4*1*1 = 64

From <https://www.sharetechnote.com/html/5G/5G_CSI_RS_Codebook.html>

Alternative options for 32 ports

Total amount of available beams

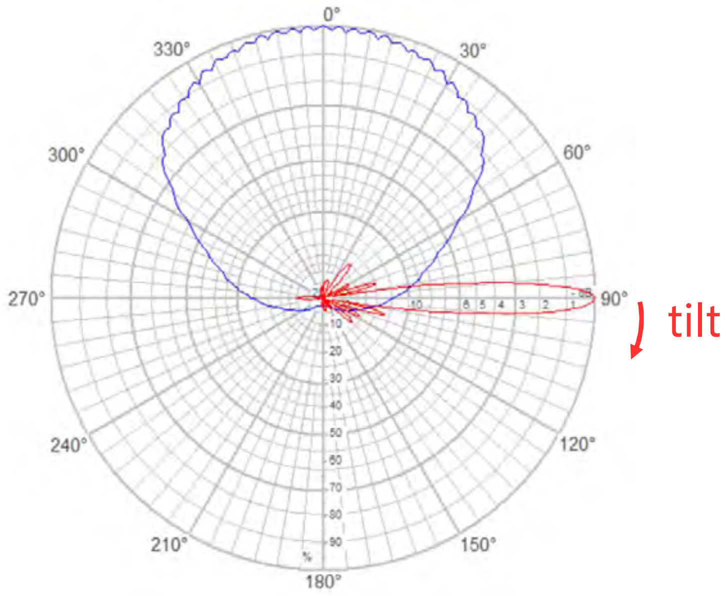
The standard allows the radio node to prevent utilization of certain beams

When multiple beams are on air power is split among beams

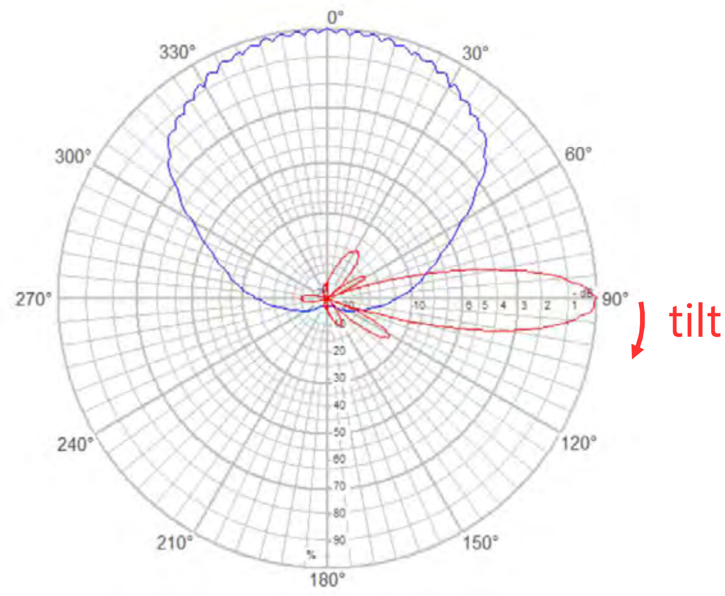
Codebook based beamforming

64T64R Antenna pattern examples

- Example of traffic beams envelope for 8 CSI-RS with 4, 1 configuration (N1, N2)
- Any tilt on the broadcast beam would have impact on the envelope of the traffic beams



- Example of traffic beams envelope for 32 CSI-RS with 8, 2 configuration (N1, N2)
- Any tilt on the broadcast beam would have impact on the envelope of the traffic beams



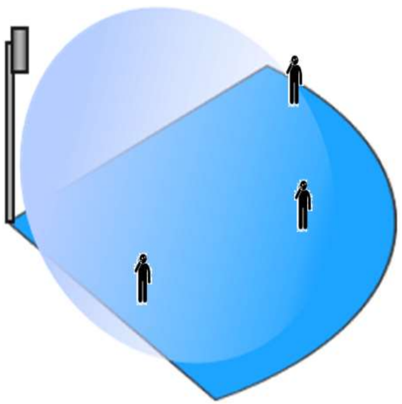
Conclusions and considerations

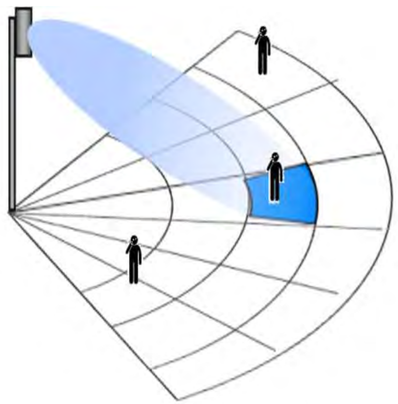
- Beamforming is possible with both passive and active antennas
- When the antenna configuration allows for beamforming the antenna patterns become dependent on SW configuration mainly (within the boundaries defined by the HW)
- It is important to decouple hardware and antenna patterns
  - the same antenna could be configured to radiate differently on different sites

- There are 2 types of beams:
  - Broadcast beams: defines the coverage area of the cell, traffic is not possible outside of the coverage area of the broadcast beams
  - Traffic beams: use to transmit user data, transmission is focused in the direction of the intended user which increases signal quality and achievable rate
  - Multiple traffic beams can be on air at the same time (MIMO), each beam has lower power because power is shared among beams

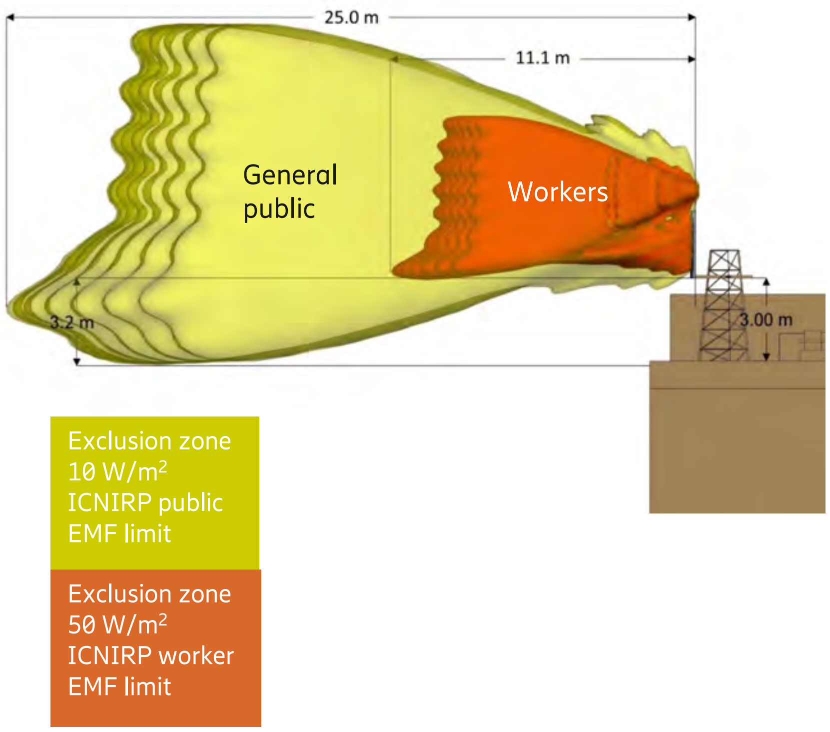
Conclusions and considerations II

- Without beamforming power is transmitted in the entire coverage area of the cell, signal to a user in the cell reaches all other users distributed in the cell
- With beamforming the power is transmitted mainly in the direction of the intended receiver, this makes the radio environment statistically less polluted which in turns leads to higher achievable bitrates





EMF Power lock



- With beamforming the instantaneous ERP and the resulting peak RF EMF levels can be higher than those for traditional base-station antennas.
- When no time-averaged power is considered the size of the exclusion zones (i.e. the areas where public access should be restricted due to RF safety limits) increases. Such increased EMF compliance boundary makes deployment challenging in e.g. dense urban environments.
- Ericsson provides a functionality that reduces the time-averaged power levels to a configurable value, this is achieved through power back-off
- Exclusion zones are reduced with no impact on coverage and minimum impact on the capacity of the served cell.

Benefits and Gains

- The EMF exclusion zones are reduced as a result of the power back-off and this is beneficial to NR deployments.
- Coverage is maintained and the impact on capacity is limited to time intervals with averaged power above the configured value.